

ShowUp Platform Documentation

Next-Generation Collegiate Event Discovery, Ticketing, UPI Payments, and Fest Management Ecosystem

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Vellore Institute of Technology (VIT), Vellore

LIVE PRODUCTION URL

[showup-campus.vercel.app ↗](https://showup-campus.vercel.app)

GITHUB REPOSITORY

[github.com/raynaagarwal01-rgb/ShowUp ↗](https://github.com/raynaagarwal01-rgb/ShowUp)

GRAVITAS '26 HUB

[/fests/gravitas26 ↗](#)

1. Executive Summary & Project Mission

At large university campuses like **VIT Vellore**, thousands of students engage in hackathons, technical masterclasses, robotics contests, and cultural festivals like **graVITas** and **Riviera**. However, event coordination has traditionally been fragmented across scattered Google Forms, WhatsApp chats, and external Unstop links.

The Solution — ShowUp: An end-to-end modern web application providing instant event discovery, team formation with secret join codes, online UPI and card payment processing, automated high-contrast offline passes (PNG photo pass), client-side 1920x1080 participation certificate generation, real-time organizer announcement broadcasts, per-event Q&A discussion forums, interactive venue destination mapping, calendar synchronization (Google & iCal), and comprehensive On-Duty (OD) attendance export for university administration.

2. Complete Page-by-Page & Button-by-Button Guide

Page 1: Landing Page (/)

The gateway welcoming students and organizers to the campus event ecosystem.

"Browse events" Button

Directs users to the live directory (/events) with full search and filters (protected by authentication).

"Host an event" Button

Navigates club leads to the Organizer Studio (/organizer) to publish new campus events.

"Sign In / Get Started" Buttons

Launches student & organizer authentication flows with immediate session persistence.

Category Pill Tiles

Instant shortcuts filtering Hackathons, Workshops, Cultural, and Competitions.

Navbar Portfolio Link

Directs logged-in students to their personal credential portfolio and bio editor (/profile).

Page 2: Events Directory Page (/events)

The centralized search and discovery engine featuring multi-dimensional filtering (protected behind student login).

Live Keyword Search Input

Real-time instant filtering matching event title, tagline, club name, venue, or college.

Scope Control ("All" / "Internal" / "External")

Filters events restricted to campus students versus open to outstation colleges.

Category Badges Row

One-click pills: "All", "Hackathon", "Workshop", "Competition", "Cultural", "Talk".

"Saved (count)" Wishlist Pill

Filters the view to only bookmarked events shortlisted by the student.

From-Date & To-Date Calendar Range Pickers

Narrows down events scheduled between specific dates.

Sort Dropdown Menu

Sorts by "Soonest First", "Latest First", "Fee: Free First", "Fee: Low to High", etc.

"Reset all filters" Button

Instantly clears all active search parameters, categories, and date boundaries.

Bookmark Ribbon (Heart/Bookmark)

Toggles immediate saving of the event to local storage without page reload.

"View on Map" / MapPin Button

Opens the destination mapping modal showing distance from campus gate and Google Maps routing.

Page 3: Event Details & Registration Page (</events/:id>)

Comprehensive event portal with interactive participation tabs and registration actions.

Tab: "Overview & Rules"

Displays complete formatted description, eligibility, and numbered contest rules.

Tab: "Announcements"

Displays real-time organizer broadcast feed with glowing URGENT ALERT badges.

Tab: "Teammate Finder"

Dedicated board where solo participants post skill requirements and connect directly via WhatsApp.

Tab: "Q&A Forum"

Allows students to submit queries and see verified responses flagged with "Answered by Organizer".

Tab: "Winners & Trophies"

Post-event podium showcasing champion teams, prize awards, and jury authentication stamps.

"Register Now" Button

Launches ShowUp Demo UPI checkout for paid events, or completes instant free registration.

"Create Team" / "Join Team" Controls

Enables collaborative group registration with 6-character cryptographic join codes.

"Download Pass (PNG)" Button

Exports the 800x1080 high-contrast door ticket to smartphone photo gallery.

"Download / Preview Certificate"

Synthesizes 1920x1080 participation credential on canvas with official gold seal.

"Add to Calendar" Dropdown

Exports to Google Calendar or downloads .ics file for Apple Calendar & Outlook.

Page 4: Dedicated graVITas '26 Fest Hub (</fests/gravitas26>)

Exclusive portal dedicated to VIT Vellore's annual international tech fest.

Live Countdown Timer

Real-time counter displaying Days, Hours, Minutes, and Seconds until Day 1 (Sep 26, 2026).

Schedule Timeline Switcher

One-click tabs: "All 3 Days (56)", "Day 1 (Fri)", "Day 2 (Sat)", "Day 3 (Sun)".

Category Filter Pills

Instant filtering across 56 synchronized VIT events (Hackathons, Workshops, Competitions).

Fest Guidelines Accordion

Covers official OD attendance, ShowUp certificates, Anna Aud/TT check-in desks, and hostel accommodation.

Page 5: Student Dashboard / My Events (</dashboard>)

Personal command center for tracking registered tickets and event materials.

"Show / Hide Ticket" Button

"Download Offline Pass" Button

Displays the full-screen scannable door QR code for rapid admission.

Saves high-res PNG pass directly to phone storage.

"Certificate" Button

Opens the verified credential modal with instant high-res PNG export.

"Add to Calendar" Dropdown

Synchronizes the event schedule with personal device calendars.

"Cancel Registration" Button (X)

Allows students to cancel attendance, automatically releasing capacity to waitlisted peers.

Page 6: Student Portfolio & Public Profile (`/profile` & `/u/:userId`)

Shareable student resume featuring verified badges and competition trophies.

"Edit Portfolio" Button

Opens the profile editor to update bio, university, grad year, skills, GitHub, and LinkedIn.

"Copy Profile Link" Button

Copies the verified public profile link to the clipboard with visual confirmation.

"Add to LinkedIn" Button

1-click URL generator pre-filling the LinkedIn Certification Form with verified ShowUp credentials.

"OD Slip" Button

Generates the official On-Duty attendance verification document.

Page 7: Organizer Studio (`/organizer`)

Mission control for university clubs and fest management teams.

"Create Event" Button

Opens the publication wizard to set title, venue, fee, schedule, and team limits.

"Manage Registrants" Button

Opens the attendee database with instant search and OD spreadsheet export.

"Door Check-in Scanner" Button

Launches the camera-based QR code ticket scanner.

"Export OD CSV" Button

Downloads 13-column academic spreadsheet formatted for the Student Welfare Office.

3. Tools & Technologies Breakdown: Who, Why & How

ShowUp is built following modern engineering best practices, leveraging industry-standard frameworks, libraries, and cloud infrastructure:

TECHNOLOGY / TOOL	WHO CREATED IT	WHY IT IS USED	HOW IT IS USED IN SHOWUP
React 19	Meta / Open Source Community	Component-driven, high-performance user interface with modern concurrent rendering.	Powers all interactive pages, modals, tab switchers, and real-time state management.
TypeScript 6	Microsoft	Static type safety; eliminates null pointer exceptions and guarantees data schema integrity.	All models (<code>EventRecord</code> , <code>Registration</code> , <code>Announcement</code> , <code>EventQuestion</code>) are strictly typed.
Vite 8	Evan You & Vite Team	Ultra-fast HMR dev server and optimized Rollup production bundler.	Compiles and bundles the application into optimized minified CSS/JS chunks in ~900ms.
Tailwind CSS v4	Tailwind Labs	Utility-first responsive CSS engine for elegant dark-mode aesthetics without bloat.	Implements custom theme palettes (<code>bg-ink</code> , <code>bg-surface</code> , <code>text-coral</code>) and mobile breakpoints.
Supabase (PostgreSQL 15)	Supabase Inc.	Cloud relational database with Row Level Security (RLS) and real-time persistence.	Stores 104+ events (including 56 graVITas events), user registrations, announcements, and Q&A.
HTML5 Canvas 2D API	W3C Web Standards	Zero-dependency client-side graphic rendering directly in the user's browser.	Synthesizes 1920x1080 participation certificates and 800x1080 offline pass PNGs offline in ~80ms.
node-qrcode	Soldair / Open Source	Fast cryptographic QR matrix generation.	Generates dynamic ticket validation QRs and UPI payment checkout QR codes.
html5-qrcode	Minhaz / Open Source	Camera-based QR code decoding engine.	Provides the live webcam scanner in the Organizer Door Check-in

TECHNOLOGY / TOOL	WHO CREATED IT	WHY IT IS USED	HOW IT IS USED IN SHOWUP
			module.
Lucide React	Lucide Project	Consistent, accessible vector iconography.	Renders all navigation, action, alert, and status icons across the entire UI.
Vercel Edge Cloud	Guillermo Rauch / Vercel	Global content delivery network (CDN) and automated Git deployment.	Hosts the live application at <code>showup-campus.vercel.app</code> with automatic SSL.
GitHub	Microsoft	Git source code version control and CI/CD triggers.	Maintains complete codebase history at <code>raynaagarwal01-rgb/ShowUp</code> .

4. Key Architectural & Functional Innovations

- **Dual-Mode Resilience (Cloud & Offline Demo Db):** If internet connection drops or API keys are missing, the system gracefully degrades to the local Indexed/LocalStorage demo database without crashing.
- **Instant Client-Side Certificate Synthesis:** Eliminates costly PDF rendering servers by generating 1080p certificates on HTML5 Canvas in under 80 milliseconds.
- **Interactive Destination Mapping & Distance Calculation:** Integrates live campus map calculations and 1-click Google Maps turn-by-turn routing for every event venue.
- **Official Academic OD Attendance CSV:** Standardized to VIT Vellore administrative guidelines for automated submission to Student Welfare Offices.
- **Comprehensive UPI Payment Gateway Simulation:** Realistic checkout modal supporting GPay, PhonePe, Paytm, and QR Scan with zero processing fees.

5. Project Verification Checklist

- ✓ **Live Production URL:** <https://showup-campus.vercel.app>
- ✓ **graVITas '26 Hub:** <https://showup-campus.vercel.app/fests/gravitas26>
- ✓ **GitHub Repo:** <https://github.com/raynaagarwal01-rgb/ShowUp>
- ✓ **Google Drive Submission Folder:**

<https://drive.google.com/drive/folders/1NqCytuNS5DiaBu40F68yNv-zc6RESz7v>

✓ **Status:** Ready for evaluation and live demonstration.